Syllabus Details

Syllabus ID:	13891
Syllabus Name:	**Basis Cross-Platform Application Programming With .NET_Lập trình ứng dụng đa nền tảng cơ bản với .NET**
Course Name English:	**Basis Cross-Platform Application Programming With .NET**
Subject Code:	**PRN212**
Learning-Teaching Method:	Social Constructivism,In-class lecture,Inquiry-based Teaching
NoCredit:	3
Degree Level:	Bachelor
Time Allocation:	Study hour (150h) = 49.5h contact hours + 145 minute final exam + 98.1h self-study
Pre-Requisite:	PRO192, DBI202
Description:	Upon completion of this course students should: 1. Understand the followings: • C# language for developing .NET applications; • Fundamental concepts of .NET Platform • Basic knowledge of Windows Presentation Foundation in .NET 2. Be able to: • Develop Cross-platform Desktop applications and support for user experience (UI & UX) 3. Be able to work in a team and present group's results
StudentTasks:	- Students must attend at least 80% of contact sessions in order to be accepted to the final examination. - Student is responsible to do all assigned exercises given by instructor in class or at home and submit on time - Use laptop in class only for learning purpose - Promptly access to the https://flm.fpt.edu.vn/ for up-to-date course information
Tools:	- Internet - Visual Studio.NET 2022 or later - MS SQL Server 2019 or later - Platform : .NET 8.0 or later - GitHub Copilot: Automated suggestions when writing code in C#, XAML, SQL, ... - AI-enhanced Debuggers (JetBrains Rider AI or Visual Studio IntelliCode) - GitHub Workflow
Scoring Scale:	10
DecisionNo MM/dd/yyyy:	1402/QĐ-ĐHFPT dated 12/12/2025
IsApproved:	**True**
Note:	
Is Scored:	**True**
MinAvgMarkToPass:	5
IsActive:	True
ApprovedDate:	12/12/2025

5 material(s)

No.	Material Description	Author	Publisher	Published Date	Edition	ISBN	Is Main Material	Is Hard Copy	Is Online	Note
1	C# 12 in a Nutshell	Joseph Albahari	O'Reilly Media, Inc.	2024		9781098147440	False	False	False	
2	https://learn.microsoft.com/en-us/ef/core/	Microsoft	Microsoft				True	False	True	
3	C# 12 and .NET 8 – Modern Cross-Platform Development Fundamentals	Mark J. Price	Packt Publishing Ltd.	2023	8th Edition	9781837635870	False	False	False	
4	https://learn.microsoft.com/en-us/dotnet/core/introduction	Microsoft	Microsoft				True	False	True	
5	https://learn.microsoft.com/en-us/dotnet/csharp/whats-new/csharp-12	Microsoft	Microsoft				True	False	True	

6 LO(s)

No.	CLO Name	CLO Details
1	CLO1	Describe about .NET Core Platform, and more new features of C# language .
2	CLO2	Demonstrate about OOP , Generic, Delegate & Event , LINQ and Design Pattern in C#
3	CLO3	Demonstrate about Windows Presentation Foundation
4	CLO4	Demonstrate about accessing to the database by Entity Framework (ORM)
5	CLO5	Explain about Concurrency programming, Stream I/O
6	CLO6	Apply Generative AI tools to support software development tasks in .NET

View mapping of CLOs to PLOs

Download All Student Material

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
1	Course Introduction Working Environment Project Introduction	Offline	CLO1	IT	Course Introduction Slide Project List	PRN212	- Choose Group (3-5 Students) - Choose Topic - Collect requirements about Topic	
2	Chapter 01: Introduction to .NET Platform & Visual Studio.NET	Offline	CLO1	TU	https://learn.microsoft.com/en-us/dotnet/core/introduction		- Read the slide, related reference documents - Do exercises - Self practice	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
3	Chapter 01: Introduction to .NET Platform & Visual Studio.NET (cont.)	Offline	CLO1	TU	https://learn.microsoft.com/en-us/dotnet/core/whats-new/dotnet-8/overview		- Read the slide, related reference documents - Do exercises - Self practice	
4	Chapter 01: Introduction to .NET Platform & Visual Studio.NET (cont.)	Offline	CLO1	TU	https://learn.microsoft.com/en-us/dotnet/core/whats-new/dotnet-8/overview		- Read the slide, related reference documents - Do exercises - Self practice	
5	Chapter 01: Introduction to .NET Platform & Visual Studio.NET (cont.)	Offline	CLO1	TU	https://learn.microsoft.com/en-us/dotnet/core/whats-new/dotnet-8/overview		- Read the slide, related reference documents - Do exercises - Self practice	
6	Chapter 01: Introduction to .NET Platform & Visual Studio.NET (cont.)	Offline	CLO1	TU	https://learn.microsoft.com/en-us/dotnet/core/whats-new/dotnet-8/overview		- Read the slide, related reference documents - Do exercises - Self practice	
7	Chapter 02: C# Programming	Offline	CLO1	TU	Chapter 3. Controlling Flow, Converting Types, and Handling Exceptions Chapter 4. Writing, Debugging, and Testing Functions (C# 12 and .NET 8 – Modern Cross-Platform Development Fundamentals)		- Read the slide, related reference documents - Do exercises - Self practice	
8	Chapter 02: C# Programming (cont.)	Offline	CLO1	TU	Chapter 3. Controlling Flow, Converting Types, and Handling Exceptions Chapter 4. Writing, Debugging, and Testing Functions (C# 12 and .NET 8 – Modern Cross-Platform Development Fundamentals)		- Read the slide, related reference documents - Do exercises - Self practice	
9	Chapter 02: C# Programming (cont.)	Offline	CLO1	TU	Chapter 3. Controlling Flow, Converting Types, and Handling Exceptions Chapter 4. Writing, Debugging, and Testing Functions (C# 12 and .NET 8 – Modern Cross-Platform Development Fundamentals)		- Read the slide, related reference documents - Do exercises - Self practice	
10	Chapter 03: Object-Oriented Programming with C#	Offline	CLO2	TU	Chapter 5. Building Your Own Types with Object-Oriented Programming (C# 12 and .NET 8 – Modern Cross-Platform Development Fundamentals)		- Read the slide, related reference documents - Do exercises - Self practice	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
11	Chapter 03: Object-Oriented Programming with C# (cont.)	Offline	CLO2	TU	Chapter 5. Building Your Own Types with Object-Oriented Programming (C# 12 and .NET 8 – Modern Cross-Platform Development Fundamentals)		- Read the slide, related reference documents - Do exercises - Self practice	
12	Chapter 03: Object-Oriented Programming with C# (cont.)	Offline	CLO2	TU	Chapter 5. Building Your Own Types with Object-Oriented Programming (C# 12 and .NET 8 – Modern Cross-Platform Development Fundamentals)		- Read the slide, related reference documents - Do exercises - Self practice	
13	Chapter 04: Collections & Generic	Offline	CLO2	TU	https://learn.microsoft.com/vi-vn/dotnet/standard/generics/collections		- Read the slide, related reference documents - Do exercises - Self practice	
14	Chapter 04: Collections & Generic (cont.)	Offline	CLO2	TU	https://learn.microsoft.com/en-us/dotnet/csharp/language-reference/builtin-types/collections		- Read the slide, related reference documents - Do exercises - Self practice	
15	Chapter 04: Collections & Generic (cont.)	Offline	CLO2	TU	https://learn.microsoft.com/en-us/dotnet/csharp/language-reference/builtin-types/collections		- Read the slide, related reference documents - Do exercises - Self practice	
16	Chapter 05: Design Pattern in .NET	Offline	CLO2	TU	https://refactoring.guru/design-patterns/csharp https://sourcemaking.com/design_patterns/		- Read the slide, related reference documents - Do exercises - Self practice	
17	Chapter 05: Design Pattern in .NET (cont.)	Offline	CLO2	TU	https://refactoring.guru/design-patterns/csharp https://sourcemaking.com/design_patterns/		- Read the slide, related reference documents - Do exercises - Self practice	
18	Chapter 05: Design Pattern in .NET (cont.)	Offline	CLO2	TU	https://refactoring.guru/design-patterns/csharp https://sourcemaking.com/design_patterns/		- Read the slide, related reference documents - Do exercises - Self practice	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
19	Chapter 06: Delegate , Event & LINQ	Offline	CLO2	TU	https://learn.microsoft.com/en-us/dotnet/csharp/programming-guide/delegates/ https://learn.microsoft.com/vi-vn/dotnet/csharp/programming-guide/events/ https://learn.microsoft.com/en-us/dotnet/csharp/linq/		- Read the slide, related reference documents - Do exercises - Self practice	
20	Chapter 06: Delegate , Event & LINQ (cont.)	Offline	CLO2	TU	https://learn.microsoft.com/en-us/dotnet/csharp/programming-guide/delegates/ https://learn.microsoft.com/vi-vn/dotnet/csharp/programming-guide/events/ https://learn.microsoft.com/en-us/dotnet/csharp/linq/		- Read the slide, related reference documents - Do exercises - Self practice	
21	Chapter 06: Delegate , Event & LINQ (cont.)	Offline	CLO2	TU	https://learn.microsoft.com/en-us/dotnet/csharp/programming-guide/delegates/ https://learn.microsoft.com/vi-vn/dotnet/csharp/programming-guide/events/ https://learn.microsoft.com/en-us/dotnet/csharp/linq/		- Read the slide, related reference documents - Do exercises - Self practice	
22	Chapter 06: Delegate , Event & LINQ (cont.)	Offline	CLO2	TU	https://learn.microsoft.com/en-us/dotnet/csharp/programming-guide/delegates/ https://learn.microsoft.com/vi-vn/dotnet/csharp/programming-guide/events/ https://learn.microsoft.com/en-us/dotnet/csharp/linq/		- Read the slide, related reference documents - Do exercises - Self practice	
23	Chapter 06: Delegate , Event & LINQ (cont.)	Offline	CLO2	TU	https://learn.microsoft.com/en-us/dotnet/csharp/programming-guide/delegates/ https://learn.microsoft.com/vi-vn/dotnet/csharp/programming-guide/events/ https://learn.microsoft.com/en-us/dotnet/csharp/linq/ Chapter 11. Querying and Manipulating Data Using LINQ (C# 12 and .NET 8 – Modern Cross-Platform Development Fundamentals)		- Read the slide, related reference documents - Do exercises - Self practice	
24	GENAI: Applying Generative AI & Workflow CI/CD in .NET	Offline	CLO6	TU	Slide		- Read the slide	

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25	GENAI: Applying Generative AI & Workflow CI/CD in .NET (cont.)	Offline	CLO6	TU	Slide		- Read the slide	
26	GENAI: Applying Generative AI & Workflow CI/CD in .NET (cont.)	Offline	CLO6	TU	Slide		- Read the slide	
27	Progress Test 1	Offline	CLO2	TU	Exam		- Do Quiz	
28	Project Review 01	Offline	CLO2, CLO6	TU	Project List		- Gather Requirements - Use GenAI to help write UseCases and preliminary UI - Create diagrams using PlantUML - Upload diagrams + documentation to GitHub	
29	Chapter 07: Building WPF Application	Offline	CLO3	TU	https://learn.microsoft.com/en-us/dotnet/desktop/wpf/?view=netdesktop-8.0		- Read the slide, related reference documents - Do exercises - Self practice	
30	Chapter 07: Building WPF Application (cont.)	Offline	CLO3	TU	https://learn.microsoft.com/en-us/dotnet/desktop/wpf/windows/?view=netdesktop-8.0		- Read the slide, related reference documents - Do exercises - Self practice	
31	Chapter 07: Building WPF Application (cont.)	Offline	CLO3	TU	https://learn.microsoft.com/en-us/dotnet/desktop/wpf/windows/?view=netdesktop-8.1		- Read the slide, related reference documents - Do exercises - Self practice	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
32	Chapter 07: Building WPF Application (cont.)	Offline	CLO3	TU	https://learn.microsoft.com/en-us/dotnet/desktop/wpf/controls/styles-templates-overview?view=netdesktop-8.0		- Read the slide, related reference documents - Do exercises - Self practice	
33	Chapter 07: Building WPF Application (cont.)	Offline	CLO3	TU	https://learn.microsoft.com/en-us/dotnet/desktop/wpf/controls/styles-templates-overview?view=netdesktop-8.1		- Read the slide, related reference documents - Do exercises - Self practice	
34	Chapter 07: Building WPF Application (cont.)	Offline	CLO3	TU	https://learn.microsoft.com/en-us/dotnet/desktop/wpf/controls/styles-templates-overview?view=netdesktop-8.2		- Read the slide, related reference documents - Do exercises - Self practice	
35	Chapter 07: Building WPF Application (cont.)	Offline	CLO3	TU	https://learn.microsoft.com/en-us/dotnet/desktop/wpf/data/?view=netdesktop-8.0		- Read the slide, related reference documents - Do exercises - Self practice	
36	Chapter 07: Building WPF Application (cont.)	Offline	CLO3	TU	https://learn.microsoft.com/en-us/dotnet/desktop/wpf/data/?view=netdesktop-8.1		- Read the slide, related reference documents - Do exercises - Self practice	
37	Chapter 07: Building WPF Application (cont.)	Offline	CLO3	TU	https://learn.microsoft.com/en-us/dotnet/desktop/wpf/data/?view=netdesktop-8.2		- Read the slide, related reference documents - Do exercises - Self practice	
38	Lab 1	Offline	CLO1, CLO2, CLO3	TU	Lab 01		- Do exercises - Self practice	
39	Assignment 1	Offline	CLO1, CLO2, CLO3	TU	Assignment 01		- Discuss assignment problems	
40	Project Review 02	Offline	CLO1, CLO2, CL3, CLO6	TU	Project List		- WPF Project Setup - Layout and Logic Setup - Testing (Unit Test / UI Test) - Reporting and Reflecting on GenAI Usage	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
41	Chapter 08: Working with Databases Using Entity Framework Core	Offline	CLO4	TU	Chapter 10. Working with Data Using Entity Framework Core (C# 12 and .NET 8 – Modern Cross-Platform Development Fundamentals)		- Read the slide, related reference documents - Do exercises - Self practice	
42	Chapter 08: Working with Databases Using Entity Framework Core (cont.)	Offline	CLO4	TU	Chapter 10. Working with Data Using Entity Framework Core (C# 12 and .NET 8 – Modern Cross-Platform Development Fundamentals)		- Read the slide, related reference documents - Do exercises - Self practice	
43	Chapter 08: Working with Databases Using Entity Framework Core (cont.)	Offline	CLO4	TU	Chapter 10. Working with Data Using Entity Framework Core (C# 12 and .NET 8 – Modern Cross-Platform Development Fundamentals)		- Read the slide, related reference documents - Do exercises - Self practice	
44	Chapter 08: Working with Databases Using Entity Framework Core (cont.)	Offline	CLO4	TU	Chapter 10. Working with Data Using Entity Framework Core (C# 12 and .NET 8 – Modern Cross-Platform Development Fundamentals)		- Read the slide, related reference documents - Do exercises - Self practice	
45	Chapter 08: Working with Databases Using Entity Framework Core (cont.)	Offline	CLO4	TU	https://learn.microsoft.com/en-us/ef/core/		- Read the slide, related reference documents - Do exercises - Self practice	
46	Chapter 08: Working with Databases Using Entity Framework Core (cont.)	Offline	CLO4	TU	https://learn.microsoft.com/en-us/ef/core/		- Read the slide, related reference documents - Do exercises - Self practice	
47	Chapter 08: Working with Databases Using Entity Framework Core (cont.)	Offline	CLO4	TU	https://learn.microsoft.com/en-us/ef/core/		- Read the slide, related reference documents - Do exercises - Self practice	
48	Chapter 08: Working with Databases Using Entity Framework Core (cont.)	Offline	CLO4	TU	https://learn.microsoft.com/en-us/ef/core/		- Read the slide, related reference documents - Do exercises - Self practice	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
49	Chapter 08: Working with Databases Using Entity Framework Core (cont.)	Offline	CLO4	TU	https://learn.microsoft.com/en-us/ef/core/		- Read the slide, related reference documents - Do exercises - Self practice	
50	Chapter 09: Working with Files and System.IO	Offline	CLO5	TU	Chapter 9. Working with Files, Streams, and Serialization (C# 12 and .NET 8 – Modern Cross-Platform Development Fundamentals)		- Read the slide, related reference documents - Do exercises - Self practice	
51	Chapter 09: Working with Files and System.IO (cont.)	Offline	CLO5	TU	Chapter 9. Working with Files, Streams, and Serialization (C# 12 and .NET 8 – Modern Cross-Platform Development Fundamentals)		- Read the slide, related reference documents - Do exercises - Self practice	
52	Chapter 09: Working with Files and System.IO (cont.)	Offline	CLO5	TU	Chapter 9. Working with Files, Streams, and Serialization (C# 12 and .NET 8 – Modern Cross-Platform Development Fundamentals)		- Read the slide, related reference documents - Do exercises - Self practice	
53	Project Review 03	Offline	CLO1, CLO2, CLO3, CLO4, CLO5, CLO6	TU	Project List		- Application Demo - Performance Reporting - Reflection on GenAI Usage and Teamwork	
54	Chapter 10: Working with XML and JSON Serializing	Offline	CLO5	TU	https://www.json.org/json-en.html https://learn.microsoft.com/vi-vn/dotnet/standard/serialization/system-text-json/overview		- Read the slide, related reference documents - Do exercises - Self practice	
55	Lab 2	Offline	CLO3, CLO4, CLO5, CLO6	TU	Lab 02		- Do exercises - Self practice	
56	Assignment 2	Offline	CLO3, CLO4, CLO5, CLO6	TU	Assignment 02		- Discuss assignment problems	
57	Progress Test 2	Offline	CLO3, CLO4, CLO5	TU	Exam		- Do Quiz	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
58	Chapter 11: Introduction to Concurrency Programming	Offline	CLO5	TU	https://learn.microsoft.com/en-us/dotnet/standard/threading/threads-and-threading		- Read the slide, related reference documents - Do exercises - Self practice	
59	Chapter 11: Introduction to Concurrency Programming (cont.)	Offline	CLO5	TU	https://learn.microsoft.com/en-us/dotnet/standard/threading/threads-and-threading		- Read the slide, related reference documents - Do exercises - Self practice	
60	Chapter 11: Introduction to Concurrency Programming (cont.)	Offline	CLO5	TU	https://learn.microsoft.com/en-us/dotnet/standard/threading/the-managed-thread-pool		- Read the slide, related reference documents - Do exercises - Self practice	
61	Project Evaluation	Offline	CLO1, CLO2, CLO3, CLO4, CLO5, CLO6	U			Group Presentation	
62	Project Evaluation	Offline	CLO1, CLO2, CLO3, CLO4, CLO5, CLO6	U			Group Presentation	
63	Project Evaluation	Offline	CLO1, CLO2, CLO3, CLO4, CLO5, CLO6	U			Group Presentation	
64	Project Evaluation	Offline	CLO1, CLO2, CLO3, CLO4, CLO5, CLO6	U			Group Presentation	
65	Project Evaluation	Offline	CLO1, CLO2, CLO3, CLO4, CLO5, CLO6	U			Group Presentation	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
66	Project Evaluation	Offline	CLO1, CLO2, CLO3, CLO4, CLO5, CLO6	U			Group Presentation	

0 Constructive question(s)

6 assessment(s)

No.	Category	Type	Part	Weight	Completion Criteria	Duration	CLO	Question Type	No Question	Knowledge and Skill	Grading Guide	Note
1	Assignment	on-going	2	15.0%	>0	12 slots	CLO2, CLO3, CLO4, CLO5			Studied subjects in syllabus		
2	Group Project	on-going	1	20.0%	>0	20 slots	CLO2, CLO3, CLO4, CLO5, CLO6			Studied subjects in syllabus		
3	Progress test	on-going	2	10.0%	>0	30'	CLO1, CLO2, CLO3, CLO4, CLO5	Multiple choicesMarked by Computer or a suitable format	20/each	Studied subjects in syllabus		
4	Final exam	Final exam	2	55.0%		145'	All CLOs			All subjects in syllabus	By exam board Manual evaluating	TE >= 4, PE > 0 & Min Avg Mark To Pass >= 5
5	Practical Exam	Final exam	1	25.0%	>0	85'	CLO1, CLO2, CLO3, CLO4, CLO5		Manual evaluating	All subjects in syllabus	By exam board	
6	Theoretical exam	Final exam	1	30.0%	4	60'	CLO1, CLO2, CLO3, CLO4, CLO5		50	All subjects in syllabus	By exam board	